- **Branching Process**: mean and variance of the population size in the n-th generation.
  Assumption: suppose we start the process from $Z_0 = 1$ and thus $\mathbb{E}(Z_0) = 1$ and $\mathrm{Var}(Z_0) = 0$. Let μ and σ^2 be the mean and variance for the offspring distribution $\mathbf{a} = (a_0, a_1, a_2, \ldots)$

$$\mathbb{E}(Z_n) = \mathbb{E}[\mathbb{E}(Z_n|Z_{n-1})] \overset{(1)}{=} \mathbb{E}[\mu Z_{n-1}] = \mu\mathbb{E}[Z_{n-1}] = \mu^2\mathbb{E}[Z_{n-2}] = \cdots = \mu^n\mathbb{E}[Z_0] = \mu^n.$$

  Here equality (1) is because $\mathbb{E}[Z_n|Z_{n-1} = k] = \mathbb{E}\left[\sum_{i=1}^k X_{n-1,i}\right] = \mathbb{E}[X_{n-1,i}] \cdot k = \mu k$, which means $\mathbb{E}[Z_n|Z_{n-1}]$ acts in the same way as the random variable μZ_{n-1}. Now for the variance,

$$\begin{aligned}
\mathrm{Var}(Z_n) &= \mathrm{Var}[\mathbb{E}(Z_n|Z_{n-1})] + \mathbb{E}[\mathrm{Var}(Z_n|Z_{n-1})] \\
&\overset{(2)}{=} \mathrm{Var}(\mu Z_{n-1}) + \mathbb{E}(\sigma^2 Z_{n-1}) = \mu^2\mathrm{Var}(Z_{n-1}) + \sigma^2\mu^{n-1} \\
&= \mu^2[\mu^2\mathrm{Var}(Z_{n-2}) + \sigma^2\mu^{n-2}] + \sigma^2\mu^{n-1} \\
&= \mu^4\mathrm{Var}(Z_{n-2}) + \sigma^2\mu^{n-1}(1+\mu) \\
&= \cdots = \mu^{2n}\mathrm{Var}(Z_0) + \sigma^2\mu^{n-1}(1+\mu+\mu^2+\cdots+\mu^{n-1}) \\
&= \sigma^2\mu^{n-1}(1+\mu+\mu^2+\cdots+\mu^{n-1}) \qquad (\because \mathrm{Var}(Z_0) = 0).
\end{aligned}$$

  Similarly, equality (2) is because $\mathrm{Var}(Z_n|Z_{n-1} = k) = \mathrm{Var}(\sum_{i=1}^k X_{n-1,i}) = k\mathrm{Var}(X_{n-1,i}) = k\sigma^2$, which means $\mathrm{Var}(Z_n|Z_{n-1})$ is the same as the random variable $\sigma^2 Z_{n-1}$.

- Three cases (illustrations of Z_n's distribution drew in class).
  (i) $\mu < 1$ (subcritical): $\mathbb{E}Z_n = \mu^n \to 0$ and $\mathrm{Var}(Z_n) = \frac{\sigma^2\mu^{n-1}(1-\mu^n)}{1-\mu} \to 0$ as $n \to \infty$.
  (ii) $\mu = 1$ (critical): $\mathbb{E}Z_n = 1$ and $\mathrm{Var}(Z_n) = n\sigma^2 \to \infty$ as $n \to \infty$.
  (iii) $\mu > 1$ (supercritical): $\mathbb{E}Z_n \to \infty$ and $\mathrm{Var}(Z_n) \to \infty$ as $n \to \infty$.

- Another proof of the extinction probability $\mathfrak{e} = 1$ in the subcritical ($\mu < 1$) case:

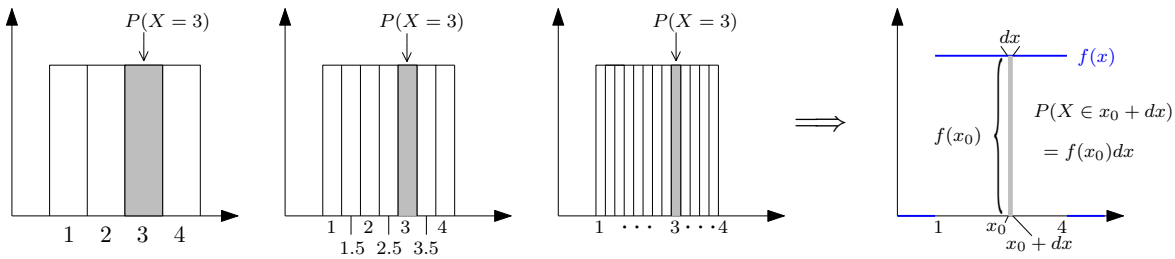
  Proof. The fact that $\mathfrak{e} = 1$ follows from

$$\begin{aligned}
1 \geq e = \lim_{n\to\infty} e_n &= \lim_{n\to\infty} P(Z_n = 0) = \lim_{n\to\infty}[1 - P(Z_n \geq 1)] \\
&= \lim_{n\to\infty}[1 - P(Z_n = 1) - P(Z_n = 2) - P(Z_n = 3) - \cdots] \\
&\geq \lim_{n\to\infty}[1 - P(Z_n = 1) - 2P(Z_n = 2) - 3P(Z_n = 3) - \cdots] \\
&= \lim_{n\to\infty}[1 - \mathbb{E}(Z_n)] = \lim_{n\to\infty}[1 - \mu^n] = 1.
\end{aligned}$$

  □

A quick review of continuous distributions

- Continuous random variables/distributions: compared with discrete random variables/distributions, the crucial relation is still to "use area to represent probability".
  For discrete distributions, we use p.m.f. $P(X = x)$ to describe the probability that a random variable takes a certain value. Pictorially, we consider each possible value is associate with a rectangle, and the rectangles are of equal width. The **area of the rectangle** corresponding to value x represents the probability $P(X = x)$. Since all rectangles are of equal width, a taller rectangle at x means the probability of taking value x is larger. See the above illustration. We normalize so that the total



area of all rectangles is **always** 1.

For continuous distributions, there are SO many values that a random variable can take. But since the total area is 1, the area of each rectangle becomes extremely small. We would necessarily have $P(X = x) = 0$ as a single line would have a zero area – for X following a continuous distribution, it does not make much sense to talk about the probability $P(X = x)$ for some value x. However, people still want to express that "it is more likely that X takes value x than y", and thus the notation "$P(X \in x_0 + dx)$" was invented: it means the probability that the random variable X is within a infinitesimal neighborhood (of length dx) near x_0 and the probability is represented using the area of the infinitesimal rectangle at x_0 with width dx. This notation is quite formal, it could mean any on of the following

$$P(x_0 \le X \le x_0 + dx), \quad P(x_0 - dx \le X \le x_0), \quad P(x_0 - dx/2 \le X \le x_0 + dx/2).$$

But since dx is infinitesimal anyway, these probabilities are the same. In fact, sometimes people simply write $P(X \in dx_0)$ when there is no ambiguity. $P(X \in x_0 + dx)$ can be calculated using the area of a rectangle, and the height of the rectangle at x_0 is called $f(x_0)$. Treating all these heights as a function of the x locations results in the so-called "density function" $f(x)$.

$$P(X \in x_0 + dx) = f(x_0)dx \quad \Leftrightarrow \quad f(x_0) = \frac{P(X \in x_0 + dx)}{dx}, \tag{$\star$}$$

i.e., $f(x_0)$ is really how much the probability changes if the value of the random variable X changes a little bit near x_0 (like a derivative). In fact, we do have

$$f(x_0) = \frac{P(X \in x_0 + dx)}{dx} = \frac{P(X \le x_0 + dx) - P(X \le x_0)}{dx} = \left.\frac{dP(X \le x)}{dx}\right|_{x=x_0}$$

$P(X \le x)$ is called the "cumulative distribution function" (c.d.f.), usually denoted by $F(x)$. It can be defined for both continuous and discrete distributions. However, the relation

$$f(x_0) = \left.\frac{dF(x)}{dx}\right|_{x=x_0} = F'(x_0)$$

only holds for continuous distributions.

Once we accept the notation ($\star$), a lot of computations for discrete random variables can be done analogously for continuous random variables, following two steps.

Step 1: change $P(X = x)$ to $P(X \in dx)$, Here, $P(X \in dx)$ means $P(X \in x + dx)$, i.e., X is in an infinitesimal neighborhood (dx) near x. This infinitesimal probability plays the same role as $P(X = x)$ in the discrete setting.

Step 2: change $\sum$ sign to $\int$ sign, if any. Here are three examples.

Example. Find $P(a < X < b)$.

$$\text{discrete}: P(a < X < b) = \sum_{x: a<x<b} P(X = x)$$

$$\text{continuous}: P(a < X < b) = \int_a^b P(X \in dx) = \int_a^b f(x)dx$$

Example. Find $\mathbb{E}(X)$.

$$\text{discrete}: \mathbb{E}(X) = \sum_x xP(X = x)$$
$$\text{continuous}: \mathbb{E}(X) = \int xP(X \in dx) = \int xf(x)dx$$

Example. Suppose $P(X > 2) \neq 0$. Find the conditional distribution of X given $X > 2$.

$$\text{discrete}: P(X = x|X > 2) = \frac{P(X = x, X > 2)}{P(X > 2)} = \begin{cases} \frac{P(X=x)}{P(X>2)}, & \text{if } x > 2 \\ 0, & \text{otherwise} \end{cases}$$
$$\text{continuous}: P(X \in dx|X > 2) = \frac{P(X \in dx, X > 2)}{P(X > 2)} = \begin{cases} \frac{P(X\in dx)}{P(X>2)}, & \text{if } x > 2 \\ 0, & \text{otherwise} \end{cases}$$

To obtain the conditional density $f(x|X > 2)$, we notice, by $(\star)$, $P(X \in dx|X > 2) = f(x|X > 2)dx$ and $P(X \in dx) = f(x)dx$. Plugging in both to the last line above, we get

$$f(x|X > 2)dx = P(X \in dx|X > 2) = \frac{f(x)dx}{P(X > 2)}, \quad \text{if } x > 2,$$

and the conditional density is $f(x|X > 2) = \begin{cases} \frac{f(x)}{P(X>2)}, & \text{if } x > 2 \\ 0, & \text{otherwise.} \end{cases}$

- Joint and conditional distribution.

$$\begin{aligned} \text{Discrete, joint p.m.f.: } P(X = x, Y = y) &= P(X = x)P(Y = y|X = x) \\ \text{Continuous: } P(X \in dx, Y \in dy) &= f_{X,Y}(x, y)dx\,dy \\ &= P(X \in dx)P(Y \in dy|X = x) \\ &= f_X(x)dx \cdot f_Y(y|X = x)dy \\ \text{which means joint p.d.f. is} \quad f_{X,Y}(x, y) &= f_X(x)f_Y(y|X = x). \end{aligned}$$

If X and Y are independent, then $P(Y = y|X = x) = P(Y = y)$ (discrete) and $f_Y(y|X = x) = f_Y(y)$ (continuous), i.e., the joint p.m.f. (discrete) or joint p.d.f. (continuous) is a product of the p.m.f. or p.d.f. for each random variable:

$$\text{if } X \text{ is independent of } Y, \text{ then} \begin{cases} P(X = x, Y = y) = P(X = x)P(Y = y), & \text{discrete} \\ f_{X,Y}(x, y) = f_X(x)f_Y(y), & \text{continous.} \end{cases}$$

Extended Reading: another application of PGF.
Consider a simple random walk $(X_n)_{n\geq 0}$ with steps $\xi_1, \xi_2, \ldots \in \{\pm 1\}$, that is,

$$X_0 = 0, \quad X_n = X_0 + \sum_{i=1}^{n} \xi_i, \quad \text{and } \xi_i's \text{ are i.i.d. with } P(\xi_i = 1) = P(\xi_i = -1) = \frac{1}{2}.$$

Let $T_1 = \min\{n : X_n = +1\}$ be the first passage time to 1. Find the p.m.f. for T_1, that is, compute $P(T_1 = k)$ for all $k \geq 1$.

First, we observe that $P(T_1 = \text{even number}) = 0$ because simple random walk can not land at $+1$ at even steps. The first a few values for $P(T_1 = k)$ are computed as follows.

$$\begin{aligned}
P(T_1 = 1) &= P(X_1 = 1) = \frac{1}{2}, \\
P(T_1 = 3) &= P(X_1 = -1, X_2 = 0, X_3 = 1) = \frac{1}{8}, \\
P(T_1 = 5) &= P\big((X_1, \ldots, X_5) = (-1, 0, -1, 0, 1) \text{ or } (-1, -2, -1, 0, 1)\big) \\
&= 2 \cdot \frac{1}{2^5} = \frac{1}{16}.
\end{aligned}$$

When T_1 equals larger values, the possible ways to get to $+1$ are a lot. The method of PGF is a quick way of getting all probabilities of the form $P(T_1 = k)$. Since T_1 only takes nonnegative integer values, its PGF exists. Call it $G_{T_1}(s)$. By definition of PGF and the law of total expectation,

$$\begin{aligned}
G_{T_1}(s) = \mathbb{E}(s^{T_1}) = \mathbb{E}[\mathbb{E}(s^{T_1}|X_1)] &= P(X_1 = 1)\mathbb{E}[s^{T_1}|X_1 = 1] + P(X_1 = -1)\mathbb{E}[s^{T_1}|X_1 = -1] \\
&= \frac{1}{2}s + \frac{1}{2}\mathbb{E}[s^{T_1}|X_1 = -1].
\end{aligned}$$

When $X_1 = -1$, the **total** number of steps T_1 to hit $+1$ for the first time will consist of the first step moving to (-1), plus a random number of steps to get to 0 from -1, plus a random number of steps to get to $(+1)$ for the first time from 0, i.e.,

$$(T_1|X_1 = -1) \quad = \quad 1 + T_{(-1)\to 0} + T_{0\to(+1)}$$

where, by shift invariance, $T_{(-1)\to 0}$ and $T_{0\to(+1)}$ are independent random variables that follow the same distribution as T_1. Thus,

$$\begin{aligned}
\mathbb{E}[s^{T_1}|X_1 = -1] &= \mathbb{E}[s^{1+T_{(-1)\to 0}+T_{0\to(+1)}}] = \mathbb{E}[s^1 s^{T_{(-1)\to 0}} s^{T_{0\to(+1)}}] \\
&= s\mathbb{E}(s^{T_{(-1)\to 0}})\mathbb{E}(s^{T_{0\to +1}}) = s\mathbb{E}(s^{T_1})\mathbb{E}(s^{T_1}) = sG_{T_1}^2(s).
\end{aligned}$$

Plugging it back, we get a *self-consistency* equation for $G(s)$:

$$G_{T_1}(s) = \frac{1}{2}s + \frac{1}{2}sG_{T_1}^2(s) \quad \Leftrightarrow \quad sG_{T_1}^2(s) - 2G_{T_1}(s) + s = 0.$$

Solving the quadratic equation, we get two roots $\frac{1\pm\sqrt{1-s^2}}{s}$. Note that $G_{T_1}(s) = \mathbb{E}(s^{T_1})$ must be nondecreasing in s on $[0, 1]$, so the PGF of T_1 should be the root:

$$G(s) = \frac{1 - \sqrt{1 - s^2}}{s}, \quad |s| \le 1.$$

Now since we know $G_{T_1}(s)$ can be computed as a power series

$$G_{T_1}(s) = \mathbb{E}(s^{T_1}) = P(T_1 = 0)s^0 + P(T_1 = 1)s + P(T_1 = 2)s^2 + \cdots,$$

we can do a Taylor expansion of the about expression in s and read out **all** the coefficients in front of the terms s^k for $k = 0, 1, 2, \ldots$, which are just the probabilities $P(T_1 = k)$. Or we can also compute the k-th derivative of $G(s)$ at $s = 0$, which is equal to $k!P(T_1 = k)$.
The Taylor expansion of $G(s)$ at 0 yields

$$G(s) = \frac{1}{2}s + \frac{1}{8}s^3 + \frac{1}{16}s^5 + \frac{5}{128}s^7 + \cdots.$$

All the s^{2k} terms vanish, because, as we discussed earlier, $P(T_1 = 2k) = 0$.